Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/26/2013
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Project Title:	Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20794	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	08/25/2011	**End Date:**	08/24/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Xiongjun (Dynaflow, Inc.) Loraine, Greg Ph.D. (Dynaflow, Inc.)		
Grant/Contract No.:	NNX11AO76A		
Performance Goal No.:			
Performance Goal Text:			
	The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse. On Earth, bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second, the tank is fixed and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50 -80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of a separator for low void fractions. We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirling flow inside the inner		

Task Description:	cylinder is generated with the help of tangential injection slots. With this configuration one can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guaranties removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. These are then removed through two two central exit orifices at each end of the cylinder. This configuration was tested successfully during a first reduced-gravity flight and we are planning additional testing this month. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in earth applications where fast separation of low void fraction gas is of concern.
Task Progress:	During this reporting period, we have continued our efforts to investigate and improve the performance of our DYNASWIRL® phase separator in both the recirculation and continuous feed operation modes. We have had the opportunity to test the DYNASWIRL® phase separator in a NASA reduced gravity flight in early May 2012 and obtained important data about its operation in reduced gravity. Systematic ground based experiments and numerical simulations were also performed to evaluate the effects of modifying various design parameters on the performance. These can be used to guide the optimization of the phase separator. Additionally, we are currently coordinating with the NASA team to prepare a test rig for a reduced gravity flight scheduled for September 2013. A. Reduced Gravity Flight Test in May 2012 An improved DYNASWIRL® phase separator was tested in reduced-gravity on May 1st and May 2nd, 2012. The test medium was a mixture of air and water. The phase separator was tested in two operation modes: a. The first was a recirculation mode, in which the mixture with reduced void fraction by the separator was fed back into the loop for further phase separation and a very low void fraction was reached. b. The second was a continuous-feed mode, in which air was continuously fed into the phase separator and removed. This was tested for void fractions up to 3% . In the recirculation mode, the gas collected in the gaseous core was extracted continuously. As the separation proceeded, an initially thick air core continued to shrink, and then eventually disintegrated as a result of strong air removal. Video recording showed that the vortex core was not affected by the gravity changes during the flight. In continuous feed mode, the mixture was produced by injecting air into the liquid flow upstream of the phase separator inlet using a Venturi air supply device sucking air from the cabin. As for the operation in recirculation mode, the pressure the operation continued unaffected by the gravitational changes, i.e. the system was operating in a balanced condition. However, fluctuations in the output were observed and depended on the changes in the chamber orientation during the parabolic flights and potentially on changes in the cabin pressure. Analysis of the test results showed that the DYNASWIRL® phase separator can successfully separate gas from the liquid under reduced-gravity and can be adapted for applications in space environment. B. Systematic Ground Based Experiments To better understand the physics of the phase separator and effects of different design parameters on the performance of phase separator, systematic ground based experiment was conducted. Flow visualization was used to observe the gaseous core formation and evolution under different operation conditions, in the meantime, relative pressure profiles in the swirl cylinder were also obtained using pressure probes which did not significantly interfere with the flow but can provide relatively fast measurements to enable evaluation of new geometry parameters. Together with measurements of flow rates and differential pressures at different test sections, the ground based experiments provide important data to evaluate the key design parameters on the performance of phase separator. Through these ground based experiments, key design parameters that affect tangential slots area and exit orifice area such as number of tangential slots, width and length of each slot, location of the slots, location of the exit orifice, number of exit orifices etc. are evaluated. Additionally, the performance of various membrane materials were tested, its air flux rate was checked against water blocking performance, and the best performed material was selected to be used as the membrane that separates the gaseous core and the gas extraction chamber, which will prevent liquid content from entering the gas extraction chamber. C. Parametric Numerical Simulations 3D Numerical simulations were conducted and involved coupling a Navier-Stokes flow solver to model the overall two phase flow field of the DYNASWIRL® phase separator and a bubble tracking and dynamics and motion solver. Both compoenents are modules of our 3DYNAFS © CFD code. The simulations enabled determination of the effects of the injection slots and of the exit orifices size and location and helped design the latest version of the separator. The simulations also highlighted the relative importance of gravity effects on the bubble capture in the separator vortex core. Bubble size effect and gravity intensity effects could be thus extracted and showed the need for zero gravity tests. D. Preparation for Upcoming Reduced Gravity Flight Test We are preparing to help NASA test the DYNASWIRL® phase separator in a reduced gravity flight scheduled for the week of September 16th 2013. We are currently working with the team of our technical monitor at NASA to integrate the phase separator into a new test rig. This integration process itself will also be a good chance to test the versatility and adaptability of the DYNASWIRL® phase separator as the NASA test rig, the operation procedures, and the orientation

	of the test section are completely different from our previous setups. Lessons learned from the integration of the DYNASWIRL® phase separator with the NASA test rig will be used to improve the adaptability of our phase separator for various applications. E. Successful Completion of the Science Concept Review During this reporting period we have successfully completed the Science Concept Review process. Recommendations from the review panel were recently received and will help us fine tune the project. We are reviewing the panel recommendations to ensure that their comments be addressed. As a result, we will improve our approach for the Requirements Definition Review (RDR) scheduled for September 2015.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Wu X, Loraine G, Chahine G. "Reduced Gravity Tests of a DynaSwirl® Phase Separator for Space Applications." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 55. , Nov-2012